

CEN/BMI 598: EMBEDDED MACHINE LEARNING

Accessible Self-Checkout: Automating Fruit Recognition

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ABSTRACT

In the realm of modern self-checkout systems within supermarkets, the manual input of fruit data emerges as a cumbersome and time-intensive process, presenting challenges for all shoppers and notably impacting individuals with visual impairments or blindness. In response to this challenge, our project harnesses the capabilities of the Raspberry Pi 3 B+ microprocessor alongside the 5MP wide-angle Raspberry Pi camera module version 1.0 to devise an innovative and inclusive self-checkout solution.

By incorporating the Raspberry Pi camera module, our system adeptly captures high-quality images of fruits, facilitating object and image identification. This image recognition process seamlessly integrates into the workflow of the self-checkout counter, alleviating the need for manual data input during the checkout process.

The crux of our project lies not only in streamlining the self-checkout experience for all shoppers but also in extending these benefits to individuals with visual impairments or blindness. Through leveraging advanced image recognition technologies, our solution fosters a more inclusive, efficient, and user-friendly retail environment. The intersection of cutting-edge hardware and software in our project represents a significant leap toward creating an accessible and equitable shopping experience for individuals of diverse abilities..

1. INTRODUCTION

Self-checkout counters have become an integral part of the modern supermarket experience, offering shoppers the convenience of a faster and more autonomous transaction process. However, the manual input of product information, especially when dealing with items such as fresh produce, can often be a daunting and time-consuming task. This challenge is exacerbated for individuals with visual impairments or blindness, who rely on assistance or costly specialized services to navigate the self-checkout process. In response to these concerns, our project introduces an innovative and inclusive solution that redefines the self-checkout experience.

The heart of our project is the utilization of cutting-edge technology, primarily the Raspberry pi 3b+ microprocessor and the Raspberry camera module V1.0, which seamlessly integrate into existing self-checkout systems. Furthermore, the integration of the Raspberry Pi camera module allows our system to capture high-resolution images of the fruit. Leveraging image classification and processing capabilities, our solution ensures accurate fruit identification, reducing the need for manual data entry and minimizing errors in the self-checkout process.

In keeping with our commitment to inclusivity, we have designed a user-friendly interface that can adapt to individual preferences. Shoppers have the option of receiving information through audio output or a visual display on an external device, making the self-checkout process accessible to everyone, regardless of their visual abilities.

In this context, our project not only streamlines the self-checkout experience for all shoppers but also extends its benefits to those with visual impairments or blindness. By providing an efficient, reliable, and inclusive self-checkout solution, we aim to create a more user-friendly and accessible retail environment, where technology serves as a bridge to a more convenient and inclusive shopping experience.

2. RELATED WORK

Recent developments collectively aim to create more accurate, efficient, and user-friendly fruit and vegetable recognition solutions for self-checkout systems. As technology continues to evolve, the retail industry can expect further enhancements in this critical aspect of automated shopping.

One such example is Vynamic Smart Vision. Diebold Nixdorf’s Vynamic Smart Vision integrates SeeWare machine learning to recognize products without barcodes, including fruits, vegetables, and baked goods. This technology will be showcased at NRF’s Big Show 2023 event in New York City. KanduAI’s Produce Recognition Technology eliminates the need for manually selecting non-barcoded items, particularly relevant for products like fresh produce. Their technology aims to enhance efficiency, improve security, and elevate the overall shopping experience. Retailers are encouraged to consider upgrading to these solutions to embrace the future of retail.

A potential solution to fully automate checkout without the need for barcodes relies on advanced machine image-recognition technology. While image recognition for pre-packaged products has seen significant progress, detecting fresh produce within semi-transparent plastic packaging poses a greater challenge due to the difficulty in obtaining accurate images through the plastic.

In the ”Optimization of existing MatchX Technology” project conducted for ITAB, a manufacturer of self-checkout technology, experts from Wageningen University Research addressed this challenge. They employed near-infrared hyperspectral imaging in combination with spectral orthogonalization, a unique data-processing technique that effectively eliminates the interference of plastic packaging from the product images. This innovative approach was successfully tested and validated for three commonly consumed fruits: bananas, apples, and pears. The equipment demonstrated its capability to identify these fruits and accurately count them, even when they were enclosed in three different types of semi-transparent packaging commonly found in Dutch supermarkets—namely, low-density, intermediate-density, and high-density polyethylene.

3. SYSTEM DESIGN

System Design:

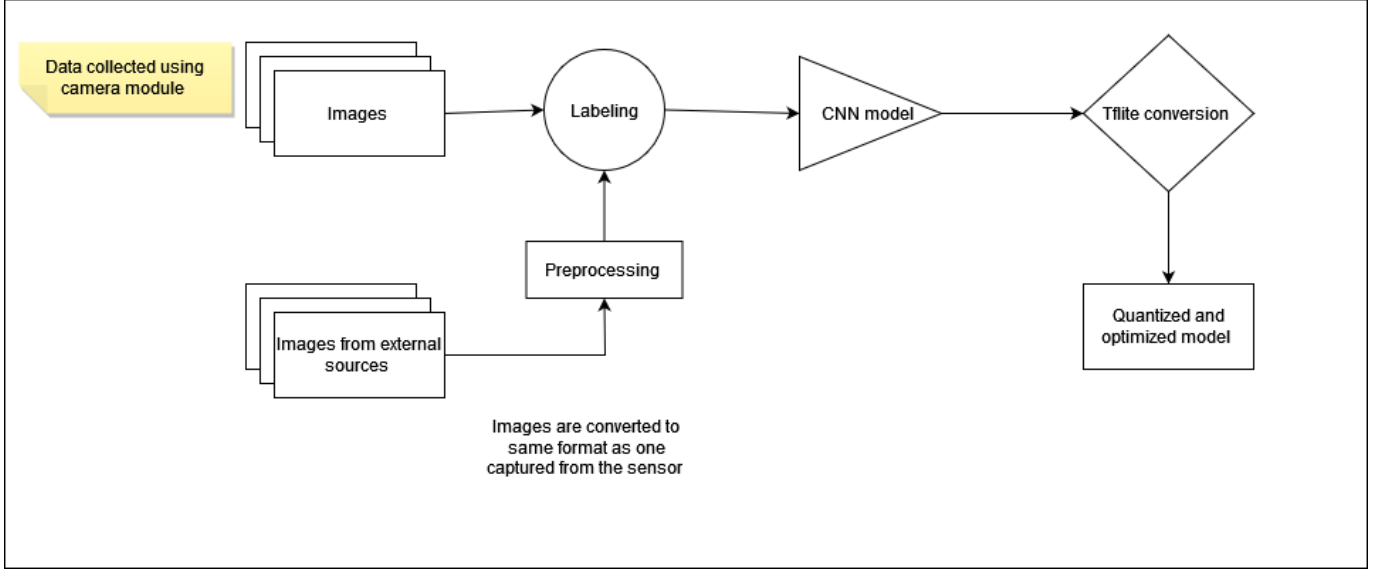


Figure 1: System design for training the machine learning model

The system design of our automated fruit recognition involves the seamless integration of hardware components, sensor technology, and machine learning algorithms to create an efficient and inclusive self-checkout experience.

Hardware Components:

- **Raspberry Pi 3b+:**

The Raspberry Pi 3B+ is a compact and versatile single-board computer designed to provide a cost-effective platform for various computing projects. Powered by a quad-core ARM Cortex-A53 processor running at 1.4 GHz, coupled with 1GB of RAM, the Raspberry Pi 3B+ offers improved processing power and multitasking capabilities compared to its predecessors. Equipped with a set of GPIO (General Purpose Input/Output) pins, HDMI output, Ethernet port, four USB ports, and built-in Wi-Fi and Bluetooth capabilities, it facilitates seamless connectivity and interaction with a wide range of peripherals. Its support for microSD cards as the primary storage medium allows for scalable and customizable storage solutions. The

Raspberry Pi 3B+ is commonly used in projects ranging from educational programming and hobbyist tinkering to more complex applications such as home automation, media centers, and Internet of Things (IoT) devices. Its open-source nature and a vibrant community contribute to its popularity, making it an accessible and powerful tool for enthusiasts and professionals alike.

- **Raspberry Pi 3b+ Camera Module:**

The Raspberry Pi Camera Module v1.0 is a compact imaging solution designed specifically for Raspberry Pi single-board computers. Developed by the Raspberry Pi Foundation, this camera module features a 5-megapixel fixed-focus camera sensor capable of capturing still images and video. The module connects to the Raspberry Pi through a dedicated camera interface (CSI) ribbon cable, ensuring a direct and efficient connection.

The camera module v1.0 has dimensions of 25mm x 20mm x 9mm, making it a small and lightweight accessory suitable for a wide range of applications. It supports video resolutions up to 1080p at 30 frames per second, providing clear and detailed footage. The fixed-focus lens simplifies usage, eliminating the need for manual adjustments, and the module is equipped with a small PCB with a camera sensor, making it easy to integrate into various projects.

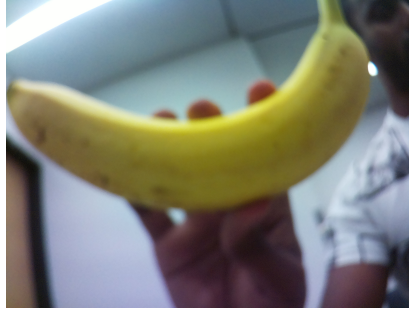
Common applications of the Raspberry Pi Camera Module v1.0 include DIY photography and videography projects, surveillance systems, and educational purposes. The module is compatible with a variety of software and libraries, including the official Raspberry Pi Camera software, as well as popular platforms like OpenCV for image processing. Its affordability, ease of use, and direct integration with Raspberry Pi boards make it a popular choice for hobbyists and developers seeking a compact and capable imaging solution for their projects.

Data Acquisition and Preprocessing:

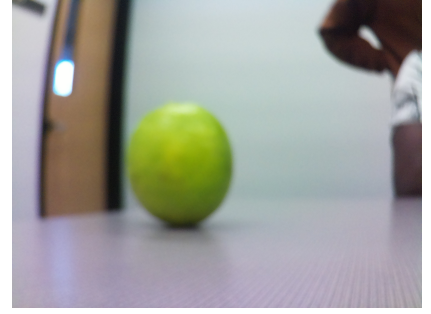
The Raspberry Pi 3b+ collects data from the camera module mounted on it. This data is crucial for obtaining metadata that enhances the identification process.



(a) Image for a Orange



(b) Image of an Banana



(c) Image of a Lime

Figure 2: Images collected through the camera module

About more than 200 samples of images of each fruit was collected using the camera module in different environments. Extra data was collected from online scrapping of relevant images which were augmented to our image data which was collected from the micro processor and camera module.

Convolution Neural Network Model Architecture

For image identification, our system utilizes a Convolution Neural Network Model architecture tailored for Raspberry Pi 3b+. While exact details may vary based on the specific machine learning model chosen, the architecture encompasses training the model on a diverse dataset of fruit images, ensuring adaptability to various fruit types and appearances.

The machine learning model is optimized for real-time execution on the Raspberry Pi 3b+, considering the device's computational constraints. Model optimization techniques, such as quantization and compression, may be employed to achieve a balance between accuracy and speed.

From previous work on convolutional neural networks for vision applications we see that depth wise separable convolution are light weight on memory and can be used in light weight embedded systems. The depth wise separable convolution are a subset of the convolution operation where the convolution operation consists of reduced number of parameters. This reduction in parameters allows the neural network to be able to deploy on light weight devices.

User Interface:

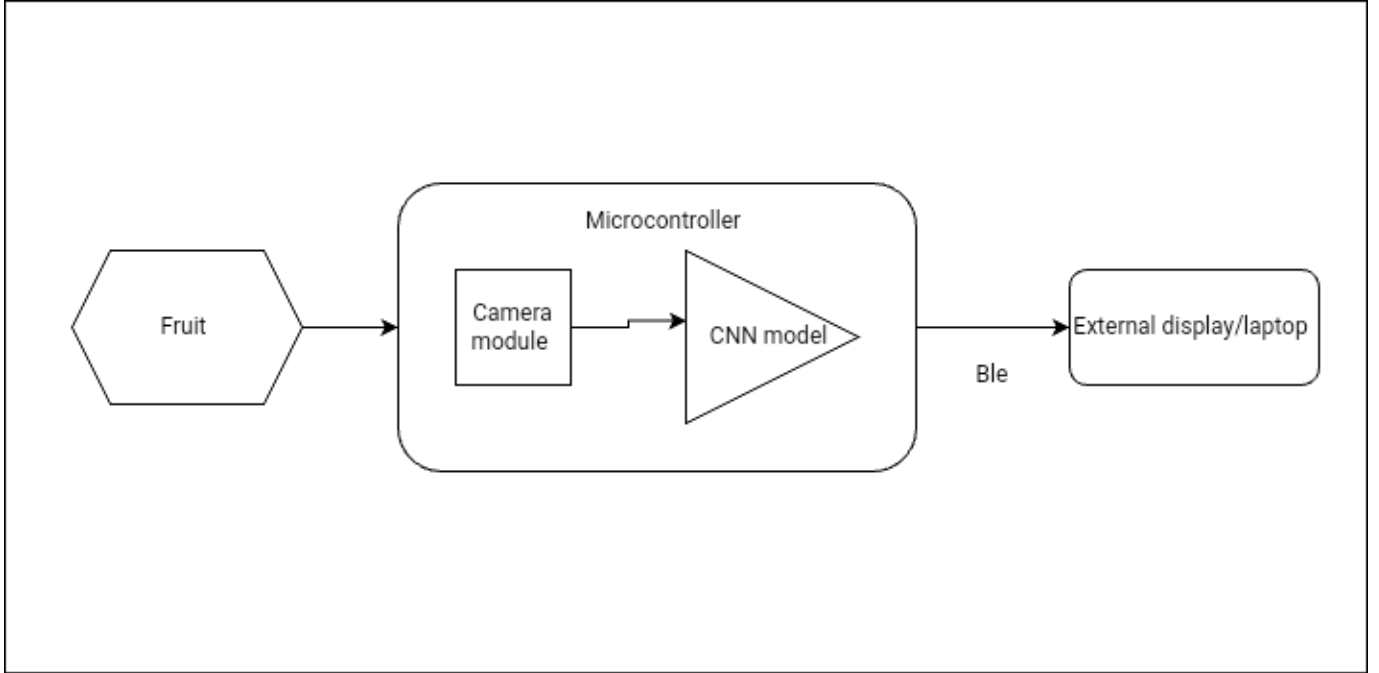


Figure 3: System design for Real-Time inference of the machine learning model on Microprocessor

System Workflow:

When a shopper places a fruit on the self-checkout counter, the Raspberry camera module v1.0 captures an image of the fruit. Simultaneously, the Raspberry Pi takes in the trained model and runs the real time inference and give the predicted class output on the terminal..

Real-Time Execution and Model Optimization:

Ensuring real-time execution is paramount for a smooth self-checkout process. Our system incorporates model optimization techniques to tailor the machine learning model for efficient execution on the Raspberry Pi 3b+. This may involve trade-offs between model accuracy and computational efficiency to meet the real-time constraints of the self-checkout environment.

The comprehensive system design presented here integrates hardware, machine learning, and other components to address the challenges of fruit identification at self-checkout counters. Through the thoughtful combination of these elements, our solution aims to provide an accessible, efficient, and inclusive self-checkout experience for shoppers of all abilities.

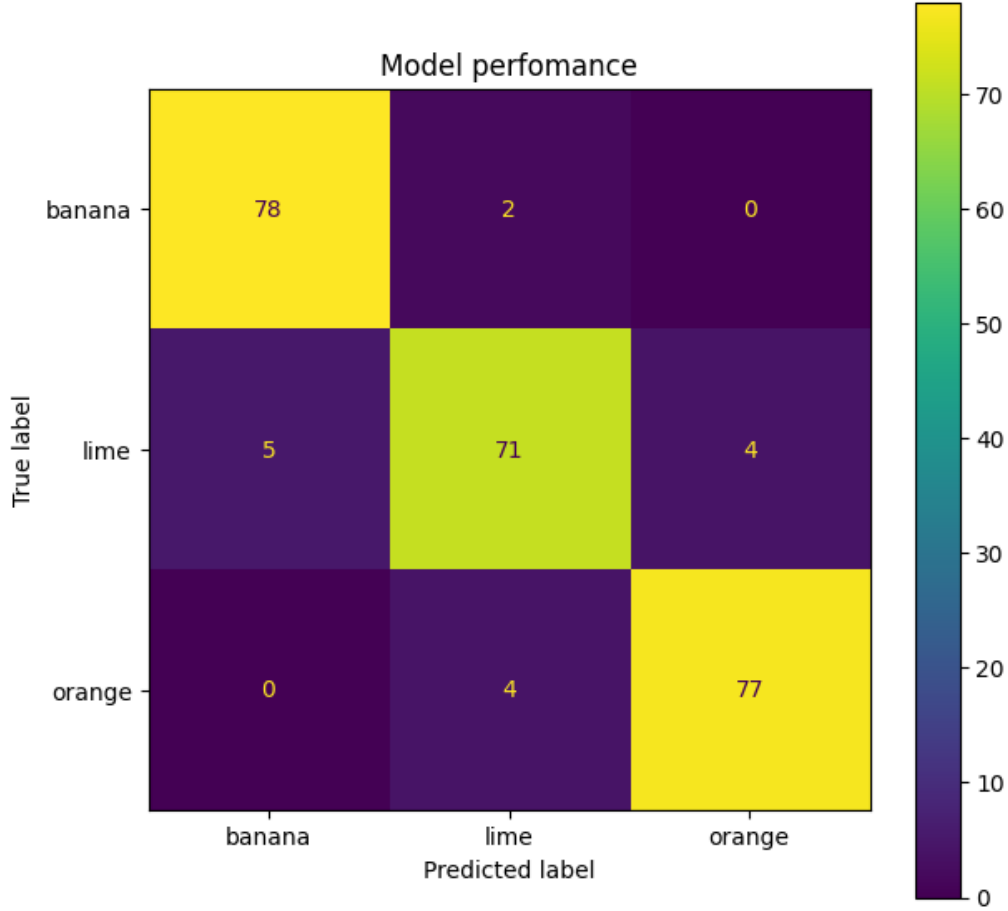


Figure 4: Confusion matrix of the test set

4. EVALUATION APPROACH

Our comprehensive evaluation approach adopts a systematic analysis framework to assess the effectiveness of automated fruit recognition. The primary objectives of our evaluation strategy include quantifying accuracy, efficiency, and inclusivity of the implemented system. The evaluation process is meticulously structured to encompass various critical aspects, covering data collection, model performance metrics, and considerations for user accessibility.

The first phase of our evaluation involves an offline assessment conducted during the training of the image recognition model. In this stage, we subject image samples to rigorous testing outside the embedded system environment. By evaluating the model’s performance on diverse datasets, we aim to gauge its accuracy, generalization capabilities, and resilience to variations in fruit appearances.

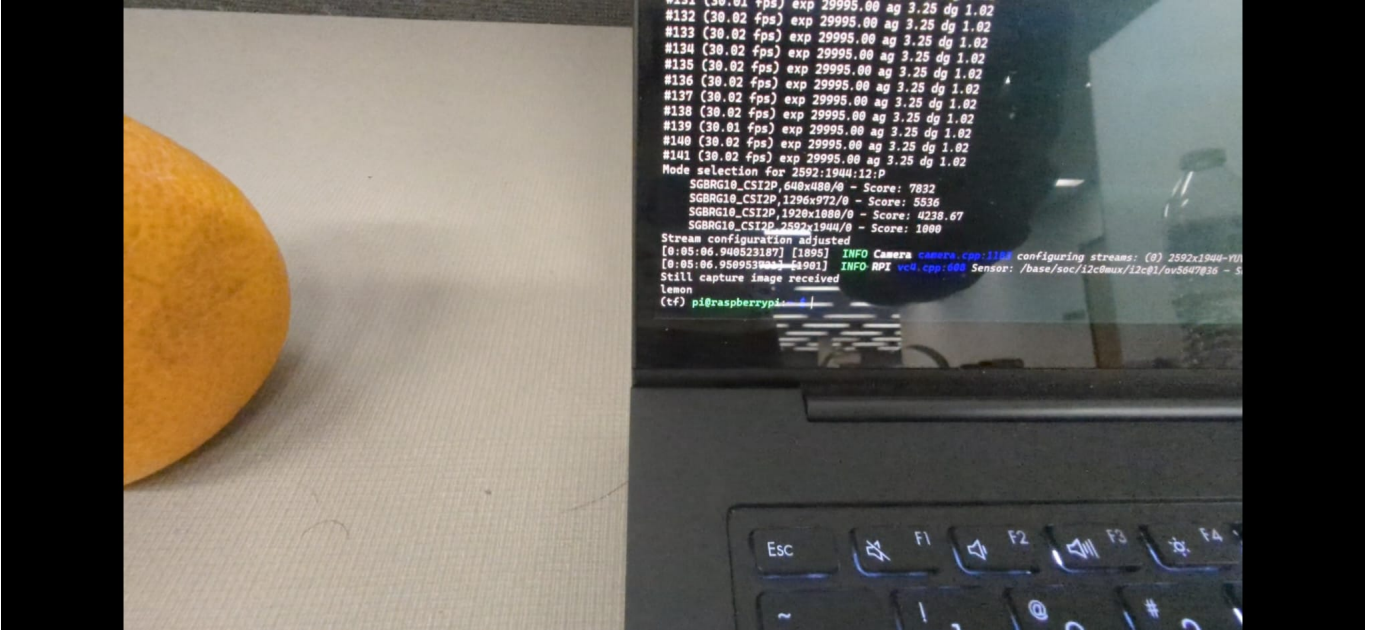


Figure 5: model output when inferred from raspberry pi

This offline evaluation provides insights into the model’s foundational robustness and its potential to handle real-world scenarios

The second phase, an online evaluation, takes place after the integration of all components into the microprocessor. This phase aims to validate the model’s performance in real-time scenarios within the embedded system. By presenting recognized objects in front of the Raspberry Pi camera module, we assess the accuracy of predictions in a dynamic environment. This process simulates real-world self-checkout scenarios, ensuring the reliability and practical applicability of the automated fruit recognition system.

To ensure the robustness and adaptability of the model, the online evaluation incorporates experiments in multiple environments. By exposing the system to diverse background settings, and fruit arrangements, we validate the model’s ability to generalize across various scenarios. This step is crucial for assessing the system’s reliability in real-world supermarket environments, where conditions may vary.

In summary, our evaluation approach combines both offline and online assessments, offering a holistic understanding of the automated fruit recognition system’s performance.